

Transport NSW Agent — Showcase Prompts

All prompts below are verified against live data. Each section lists what tool(s) fire and what the UI renders.

Commuter Queries

Next trains — SVCROW cards

What are the next 4 trains from Strathfield to Central right now?

Fires `get_next_services` (auto). Renders visual departure cards with times and platform.

Live departures board

What's leaving Central Station in the next 20 minutes?

Fires `get_live_departures` (auto). Returns live departure list across all platforms.

One specific line

Show me upcoming Metro trains from Chatswood heading into the city.

Fires `get_next_services` (metro). Filters to Chatswood → city direction SVCROW cards.

Enthusiast Queries

Fleet breakdown — what's actually running

What train types are running on Sydney Trains today? Break it down by fleet type and how many

Fires `run_sql` on `analysis.agent_train_formation` grouped by `vehicle_category_name`. Expect Waratah (8-car), Tangara, Oscar, Millennium. Should render a bar chart.

Peak frequency deep-dive

How often does the M1 Metro run at Chatswood through the day? When are trains most frequent

Fires `run_sql` on `analysis.agent_stop_frequency` (metro, Chatswood). Headways drop to 4 min at peak (7–9am), 9–10 min off-peak. Renders a line chart.

Which line dominates the timetable

Which Sydney Trains T-lines run the most scheduled trips in the whole timetable? Is T1 really

Fires `run_sql` on `analysis.agent_route_summary` grouped by `route_name`. T1 = 11,868 trips total, ahead of T8, T2. Bar chart expected.

Multi-Dimensional Queries

Cross-network live comparison

How many trains and metros are live on the network right now? Compare Sydney Trains vs Metro which has more vehicles out?

Fires `get_vehicle_position` (auto, limit=300). Returns 268 Sydney Trains vs 32 Metro live. Agent summarises both networks side by side.

Hourly service volume at a major station

Show me a bar chart of how many train services pass through Central Station each hour on a weekday

Fires `run_sql` on `analysis.agent_stop_frequency` (sydneytrains, Central, weekday) grouped by hour. Peak around 7-9am and 5-6pm. Renders a bar chart.

Crush vs comfort across T-lines

Compare T1, T2, T4, T8 and T9 - which line has the most crush-loaded departures in the schedule

Fires `run_sql` on `analysis.agent_occupancy_advisory`. T1 has 261 crush-load departures vs near-zero for T4. Multi-series bar chart.

Operational Queries

Network interchange hubs

Which stations on the Sydney Trains network are the biggest transfer hubs? Show me the top 8

Fires `run_sql` on `analysis.agent_transfer_hubs` (deduplicated by station). Redfern (30 routes, score 65.3), Central (34 routes), Lidcombe, Glenfield. Bar chart by score.

Live service alerts

Are there any active service disruptions or alerts on Sydney Trains right now?

Fires `get_active_alerts` (sydneytrains). Live alert exists: Macquarie Fields station access changes. Returns cause, effect, and description text.

Delay snapshot — who is running late

Which Metro trips are running with the most delay right now? Give me the worst offenders.

Fires `get_worst_delays` (metro, 2hr). Returns peak delay in minutes per trip, first/last seen time, and snapshot count.

Network delay trend

Plot the average delay trend across Metro and Sydney Trains over the last 2 hours as a line

Fires `get_delay_trend` (auto, 2hr). Returns 5-min buckets with avg/max delay. Renders a line chart with time on X-axis.

Silly Queries

The marathon train

What's the longest train journey in the entire Sydney Trains timetable? How long does it take?

Fires `run_sql` on `analysis.agent_trip_summary` ORDER BY `run_time_minutes` DESC. Top result: Central → Brisbane Roma St, 850 minutes (14h 10m), 83 stops. Unexpected result that lands well.

Rush hour dodge

If I'm trying to avoid being sardined on a train, which lines and times should I absolutely avoid?

Fires `run_sql` on `analysis.agent_occupancy_advisory` filtering for `crushed_standing` between 7–9am. Returns BMT and T1 routes at specific stations.

Do trains actually run at 3am?

Are there any trains running at 3am between Parramatta and the city, or am I stuck getting a taxi?

Fires `get_next_services` (from=Parramatta, to=Central, after_time=03:00). Will find night owl services or correctly report no services.

Chart Queries

Bar Chart

Show me a bar chart of the total number of scheduled trips per Sydney Trains T-line in the t

run_sql → *agent_route_summary* GROUP BY route_name WHERE T-lines
only → *render_chart* bar. Verified: T1 top, T3/T5/T6 at bottom.

Doughnut / Pie

Give me a doughnut chart of Metro occupancy levels across all live trains right now - how many are empty, few seats, or standing room only?

run_sql → *agent_live_vehicle_state* GROUP BY occupancy_status
(metro) → *render_chart* doughnut. Verified: STANDING_ROOM_ONLY=15,
FEW_SEATS=9, MANY_SEATS=9.

Line Chart — Headway over the day

Plot Metro service frequency at Chatswood through the day as a line chart - show me how headway in minutes changes from midnight to midnight on a weekday.

run_sql → *agent_stop_frequency* (metro, Chatswood, weekday) GROUP BY
hour → *render_chart* line. Peak drops to 4 min, late night stretches to 10 min.

Line Chart — Delay Trend

Show me a line chart of average train delays over the last hour across the whole network. Is

get_delay_trend (auto, 1hr) → *render_chart* line. X = time buckets, Y =
avg delay seconds. Metro delay visible from recorded history.

Quick-fire Sequence (for live demo flow)

Run these in order for a smooth 5-minute demo:

1. What's leaving Central Station in the next 15 minutes? — live departures card
2. Show me a bar chart of scheduled trips per T-line. — bar chart
3. How many trains are live on the network right now vs Metro? — cross-network
4. Plot the delay trend over the last 2 hours as a line chart. — live trend

5. Give me a doughnut of Metro occupancy levels right now. — live pie
 6. What's the longest train journey in the timetable? — silly reveal
 7. Are there any service alerts on Sydney Trains today? — live alerts
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Notes for Demo

- **SVCROW cards** only render for `get_next_services` and `get_live_departures` — those always produce the visual card UI
- **Charts** render in the map overlay; clicking closes and returns to map view
- **Live data** (vehicles, delays, alerts) requires `test_realtime.py` running — confirm it is up before demoing queries 4–5
- `run_sql` queries default to `metro` DB unless `transport_type` is specified in the prompt or inferred — for T-line queries say “Sydney Trains” explicitly to be safe